Learning Week: Day 1 → Day 6

A complete guide to understanding chains, agents, memory, orchestration, and building a real retail data pipeline

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Day 1: Chains | Day 2: Templates | Day 3: Memory & Tools | Day 4: LangGraph | Day 5: Orchestrator | Day 6: The Project

Day 3: Memory & Tool-Calling

Conversation Context & External Functions

Memory: Giving LLMs Context

By default, an LLM has no memory — each call is independent. It can't remember previous messages or refer to earlier parts of a conversation.

WITHOUT MEMORY:

```
User: "My name is Ahmed"
Bot: "Nice to meet you, Ahmed!"

User: "What's my name?"
Bot: "I don't know, you didn't tell me" ✗
```

WITH MEMORY:

```
memory = ConversationMemory()
# Keeps track of all messages
# Feeds them back into each new call
memory.add("User: My name is Ahmed")
memory.add("Bot: Nice to meet you, Ahmed!")

User: "What's my name?"
Bot: "Your name is Ahmed" ✓
```

Tool-Calling: Giving LLMs Actions

Tool-calling means giving the LLM access to functions it can choose to use — a calculator, a search engine, a database lookup. The LLM decides **when** to call a tool and **what** to pass into it.

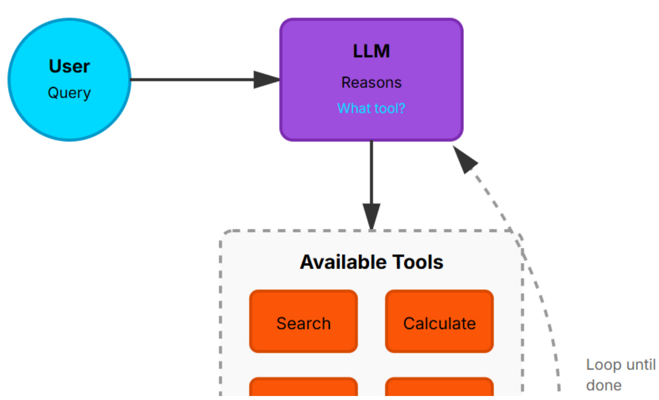
WITHOUT TOOLS:

```
User: "What's 2847 × 3956?"
LLM: "I think it's around 11 million" ✗
```

WITH CALCULATOR TOOL:

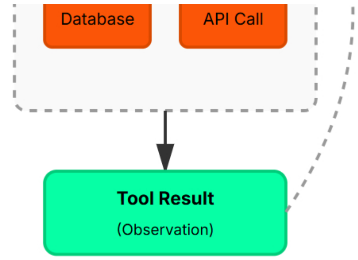
```
Tools available: [calculator, search, database]
User: "What's 2847 × 3956?"
LLM: "I'll use the calculator tool"
Result: "11,270,632" ✓
```

How Tool-Calling Works (ReAct Loop)



ReAct Loop:

1. Reason
2. Act (pick tool)
3. Observe result
4. Repeat if needed
5. Return answer



Key Concepts

Conversation Memory: Stores chat history and feeds it back

Chat History: List of past user/assistant messages

Tool/Function Calling: LLM decides which tool to use

Tool Schema: Definition of a tool (name, description, parameters)

ReAct Loop: Reason → Act → Observe → Repeat

Real-World Example

Query: "What's the weather in Paris, and what should I wear?"

1. LLM thinks: "I need current weather data"
2. Calls weather tool → Gets "10°C, rainy"
3. LLM thinks: "Now I have weather, I can answer"
4. Final answer: "Bring an umbrella and a warm jacket"

Key Takeaway: Memory lets LLMs remember context. Tool-calling lets them take actions. Together, they transform a simple language model into an intelligent agent that can hold conversations and get real work done.

Free Resources to Dive Deeper

LangChain Docs
Official documentation with API reference and guides

LangChain Academy
Free official courses on LangChain fundamentals

LangGraph Docs
Guide to building graphs and agents

Multi-Agent Tutorial
Real example of orchestrator agents in action